



Fig. 4: Installation of Python-dev



Fig. 5: Rpi-GPIO installation



Fig. 6: Command for viewing source code in nano editor



Fig. 7: Running the code



Fig. 8: Setting the date and time

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download on www.efymag.com website.

GPIO pins, I²C interface, SPI interface, serial Tx/Rx pins and PWM pins that can be used to control the external hardware.

Fig. 3 shows connection of Raspberry Pi's GPIO to the 16x2 LCD for displaying the output. Only four data pins are used for writing on the LCD panel. Either the first four or the last four data pins are to be used. In our case, the second set of data pins (pin D4, D5, D6 and D7) are used.

Working of the project is simple. When you switch on the Raspberry Pi board, it starts booting up and fires the small Python program. The program determines the IP address of Raspberry Pi and writes it down on the first line of the LCD panel. Once the IP address is known, you can easily access it using SSH. The second line of the LCD shows the current time.

Software

First, you have to install Python development toolkit (required for RPi.GPIO) using the following command (see Fig. 4):

```
$ sudo apt-get
install python-dev
```

Similarly, install Python libraries by writing the following command in the terminal as shown in Fig. 5:

```
$ sudo apt-get
install python-rpi.
gpio
```

After installations, download the source code (iptalk.py) and move it to a directory of your choice. You can see the source code in nano text editor using the following command:

```
$ sudo nano iptalk.py
```

Now you can run the code using the below-mentioned command as shown in Fig. 7:

```
$ sudo python
iptalk.py
```

You can set the date and time by writing the following command as shown in Fig. 8:

```
sudo date --set "23 APR 2012 16:01"
```

Now the last task is to make this work on every boot. For that, open 'etc/rc.local' file using the below command:

```
$ sudo nano /etc/rc.local
```

Copy the following line in the beginning of this file:

```
sudo python /where-your-file-is/
iptalk.py
```

Save it (Ctrl+o), exit (Ctrl+x) and reboot your Raspberry Pi. Now every time you boot your Raspberry Pi, it will show the IP address of the device along with the time. ●

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